Politecnico di Torino

Nuclear fission plants

Assignment 1:

Analysis of a Passive Decay Heat Removal System

Master degree in Sustainable nuclear energy

Author:

Sciavolino Maria Elena s308434

Polato Riccardo s359217

Venturo Ludovica s359211

June 4, 2026

Table of Contents

Chapter 1

Assignement 1.1: Introduction to Passive Decay Heat Removal Systems mechanisms

1.1 Problem Statement

Natural circulation is a key phenomenon in nuclear engineering, particularly for the design of Passive Decay Heat Removal (PDHR) systems. The primary goal of this exercise is to model a simple natural circulation loop to understand the engineering constraints and thermal-hydraulic mechanisms that drive flow without external energy sources.

Specifically, this study analyzes a rectangular loop where the fluid moves due to buoyancy effects caused by density gradients. The loop consists of a heater that brings the water from saturated liquid to saturated steam, and a cooler that returns the steam to a saturated liquid state [1] .

1.1.1 Input data and assumptions

To model the thermal-hydraulic behavior of the loop, the following operating parameters and simplifying assumptions are adopted. The main design data are summarized in Table 1.1.

- **Steady-state**: The system is in permanent equilibrium.
- **Saturation**: Both phases are always at saturation conditions for 70 bar.

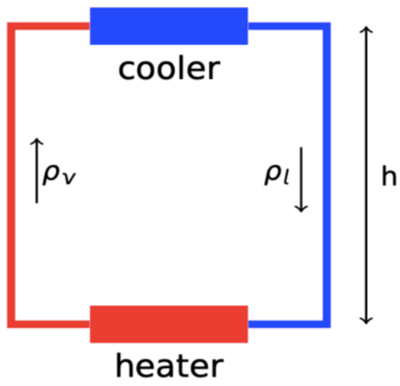

Figure 1.1: Schematic representation of the rectangular loop where the fluid flow is driven by buoyancy effects.

- **Insulation**: Heat losses to the environment are neglected.
- **Layout**: Heater and cooler are perfectly horizontal.

Table 1.1: Input parameters and design data for the natural circulation loop.

Parameter			
Thermal Power	P_{th}	34.8	MW_{th}
Operating Pressure	p	70	bar
Gravity acceleration	g	9.81	m/s^2
Reference Pipe Length	L	10	m
Pipe Schedule (ANSI B36.10)	-	STD	-
Pipe Roughness (Goal C)	ϵ	5.0×10^{-5}	m
Heater loss coefficient	k_{heater}	40	-
Cooler loss coefficient	k_{cooler}	20	-

1.1.2 Goals

The assignment is divided into three main sub-goals:

1. Find the required height h such that the total power is removed by natural circulation, given a fixed section length $L = 10\,\text{m}$.
2. Determine the configuration where the section length L is a free parameter and find the height h such that $h = L$.

3. Repeat the analysis for the condition $h = L$ considering a pipe roughness of $\epsilon = 5.0 \times 10^{-5}$ m and discuss the differences compared to the smooth pipe case.

1.2 Mathematics

1.2.1 Basic equations in conservative form

The physical behavior of the fluid within the natural circulation loop is governed by the Navier-Stokes equations, representing the conservation of mass, momentum, and energy. In their general differential form, these are expressed as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \tag{1.1}$$

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right) = -\nabla p + \nabla \cdot \underline{\underline{\tau}} + \rho \vec{g} \tag{1.2}$$

where ρ is the fluid density, $\vec{v}$ is the velocity vector, p is the pressure, $\underline{\underline{\tau}}$ is the viscous stress tensor, and $\vec{g}$ is the gravitational acceleration.

1.2.2 Simplification and final equations

To solve the problem, several simplifying assumptions are applied: steady-state ($\partial/\partial t = 0$), one-dimensional flow along the loop axis, and incompressibility within each leg. Integrating the momentum balance along the closed loop, the buoyancy driving head must balance the total pressure losses:

$$\Delta p_{buoyancy} = \Delta p_{losses} \implies (\rho_l - \rho_v) g h = \Delta p_{dist} + \Delta p_{conc} \tag{1.3}$$

where distributed losses $\Delta p_{dist} = \sum (f \frac{L}{D} \frac{1}{2} \rho v^2)$ and concentrated losses $\Delta p_{conc} = \sum (k \frac{1}{2} \rho v^2)$.

Resolution for Design Goals

The design goals require isolating specific geometric parameters from the pressure balance equation.

Goal 1: Height derivation with fixed L

In the first scenario, the length of the legs is fixed ($L = 10$ m). The required height h to maintain the flow is derived directly by dividing the total pressure losses by the hydrostatic gradient:

$$h = \frac{\sum_i \left(f_i \frac{L}{D} \frac{1}{2} \rho_i v_i^2 \right) + \sum_j \left(k_j \frac{1}{2} \rho_j v_j^2 \right)}{(\rho_l - \rho_v) g} \tag{1.4}$$

Goal 2: Critical length derivation for h = L

In the second scenario, we impose the condition $h = L$, where the vertical height of the loop matches the physical length of the vertical pipes. Substituting h with L in the balance equation:

$$(\rho_l - \rho_v) g L = L \cdot \left[\sum_i \left(\frac{f_i}{D} \frac{1}{2} \rho_i v_i^2 \right) \right] + \Delta p_{conc} \tag{1.5}$$

To find the critical length L, we group the terms proportional to L and solve:

$$L = \frac{\Delta p_{conc}}{(\rho_l - \rho_v) g - \sum_i \left(\frac{f_i}{D} \frac{1}{2} \rho_i v_i^2 \right)} \tag{1.6}$$

This formulation highlights that a physical solution exists only if the buoyancy head per unit length $(\rho_l - \rho_v) g$ is greater than the frictional pressure drop per unit length.

1.2.3 Parameters and Correlations

The system is closed using the mass flow rate from the energy balance $\dot{m} = P_{th}/(h_v - h_l)$ and the Reynolds number $Re = \rho v D/\mu$. For smooth pipes, the Blasius correlation is used: $f = 0.316/Re^{0.25}$. For Goal 3 (rough pipes), the Haaland (left) and the Colebrook (right) equations are employed:

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\left(\frac{\epsilon/D}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right] \qquad \frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\epsilon/D}{3.7} + \frac{2.51}{Re\sqrt{f}} \right) \tag{1.7}$$

1.3 Solution

The model was implemented in Python, utilizing the *CoolProp* library to ensure high-precision thermophysical properties for water in saturation conditions. To identify the optimal pipe configuration, a parametric analysis was performed across the entire range of provided commercial diameters; for each iteration, the internal diameter was converted to SI units starting from the nominal outside diameter and wall thickness (t) using the relation $D_{in} = D_{out} - 2t$.

1.3.1 Method implemented

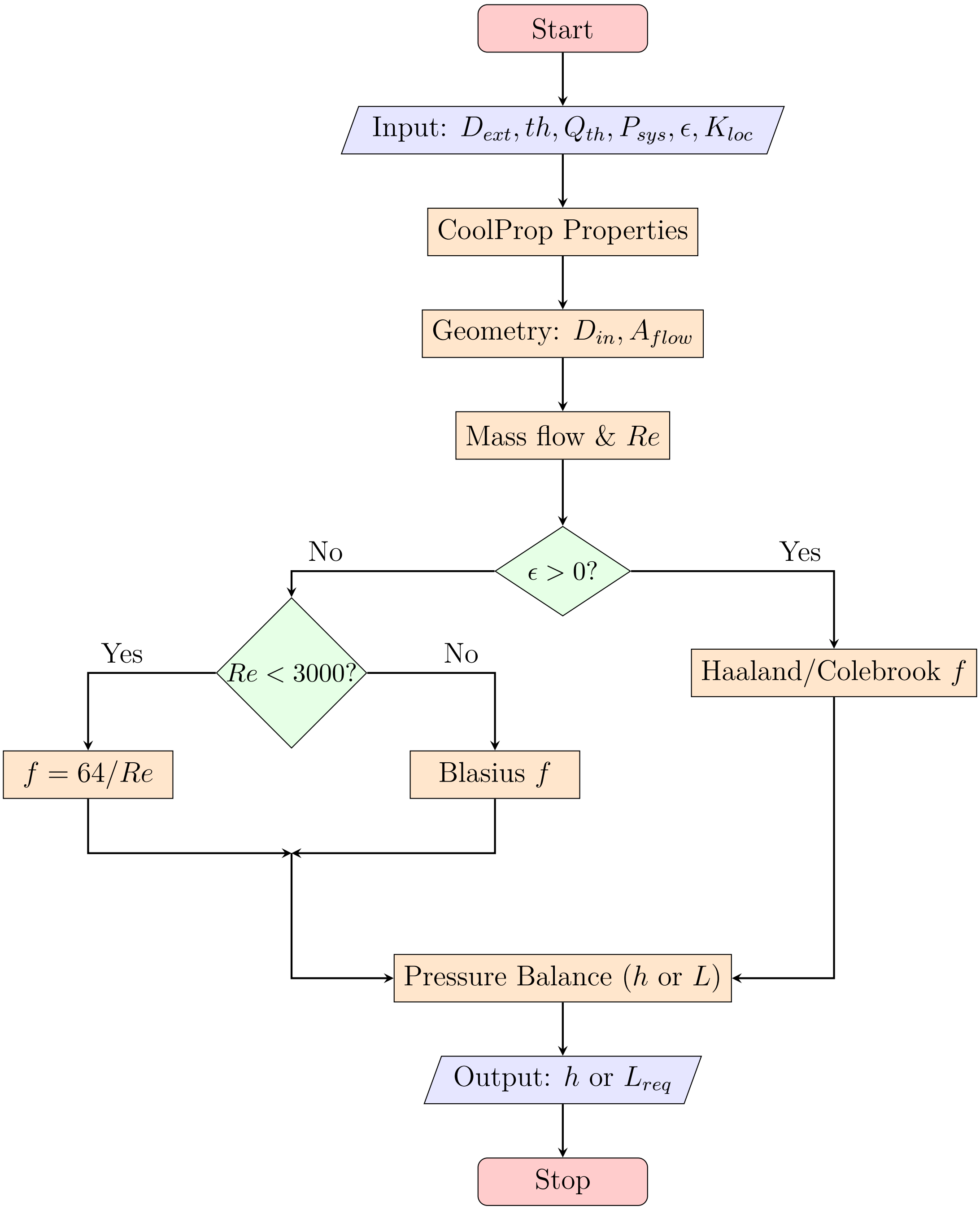

Figure 1.2: Flowchart of the algorithm.

1.3.2 Results

Both configurations analyzed impose constraints on the pipe diameter in order to obtain physically meaningful results. The following conditions must be satisfied:

- 1) the height must be less than or equal to the length of the pipe ($h < L$).
- 2–3) the calculated length must be a positive value ($L > 0$).

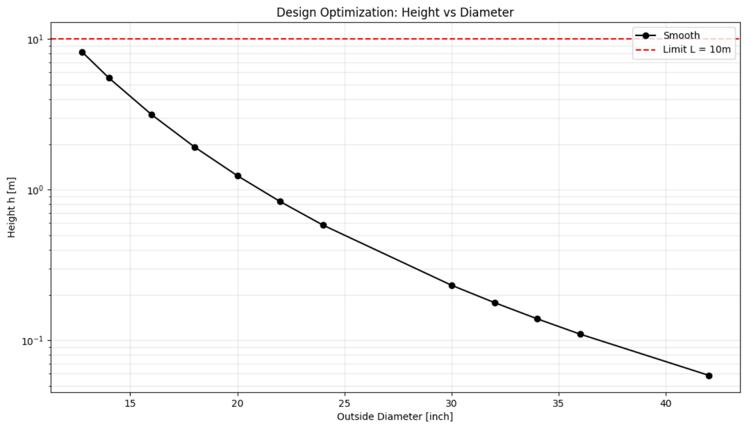

Figure 1.3: Required height h depending on outside diameter for smooth pipes ($L = 10$ m).

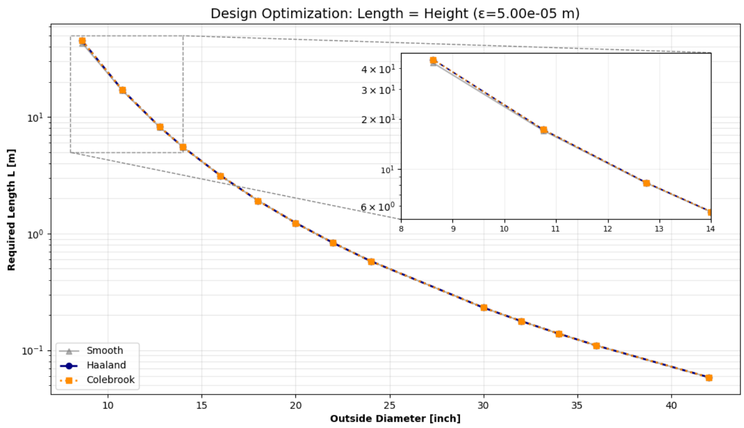

Figure 1.4: Design optimization for $L = h$ comparing smooth, Haaland, and Colebrook models ($\epsilon = 5 \cdot 10^{-5}$ m).

Case	**Condition**	**Min OD [inch]**
Goal 1	Smooth ($L = 10\,\text{m}$)	12.75
Goal 2	Smooth ($L = h$)	8.625
Goal 3	Rough ($L = h$)	8.625

Table 1.2: Summary of the minimum feasible diameters

Mathematically, there is no upper limit to the pipe diameter. However, beyond a certain threshold, the required dimensions for h and L become unrealistically small, rendering the design economically and structurally unjustifiable and unfeasible. For these reasons, only the most suitable configurations were selected, specifically those corresponding to the smallest permissible diameters. The results for these cases are presented in Table 1.3.

OD [in]	h ($L = 10m$) **[m]**	$L = h$ **Smooth [m]**	$L = h$ **Rough (H-C) [m]**
8.625	$42.072^{\dagger}$	43.145	45.201 - 45.198
10.750	$16.913^{\dagger}$	16.990	17.207 - 17.207
12.750	8.215	8.207	8.245 - 8.245
14.000	5.525	5.512	5.526 - 5.526

Table 1.3: Comparison of the selected pipe diameters. † Value exceeds the physical design limit of 10 m

It is worth noting that the results for $L = 10$ m (Point 1) and the self-sustaining condition $L = h$ (Point 2) show minimal deviation. This consistency is attributed to the dominance of localized pressure drops (K_{loc}) over distributed friction losses for the selected diameters.

Based on the thermohydraulic analysis, the 12.750 inch pipe is selected as the optimal design solution. It is the smallest commercial diameter that satisfies the primary constraint of $h \leq 10$ m. The configuration demonstrates high robustness, with less than 10% sensitivity to surface roughness ($\epsilon = 5 \cdot 10^{-5}$ m). Consequently, the 12.750" pipe represents the best compromise between hydraulic efficiency, material costs, and structural requirements for the 34.8 MW thermal load.

Text reviewed and corrected with the support of Artificial Intelligence.

Chapter 2

Assignement 1.2: Passive Decay Heat Removal Systems

The goal of this assignement is to analyse in detail a passive Decay Heat Removal System (DHRS) of a nuclear fission reactor. In this case the design is given, and the aim is to verify that the design is compliant with system requirements.

2.1 Problem Statement

The DHRS starts operating following reactor shutdown to provide core cooling and exhaust the decay heat. It can be divided in three main circuits:

- The Primary System Circuit (PSC), which removes the heat from the reactor core
- The Intermediate System Circuit (ISC), which receives heat from HX1
- The Tertiaty System Circuit (TSC), which receives heat from HX2 and transport the steam produced to a cooling tower, where the steam is condensed.

The DHRS remains in standby during nominal operation, with a check valve in the cold leg preventing flow within the Primary System Circuit (PSC). Following a primary pump trip in accidental conditions, the resulting pressure differential triggers the valve to open. This initiates natural circulation, driven by the density gradient between the vessel downcomer and the core. The heat is then sequentially transferred from the PSC to the Intermediate System Circuit (ISC) via HX1, and subsequently to the HX2 pool. Once saturation is reached in the pool, the generated steam is discharged to the cooling tower through the Tertiary System Circuit (TSC) [1].

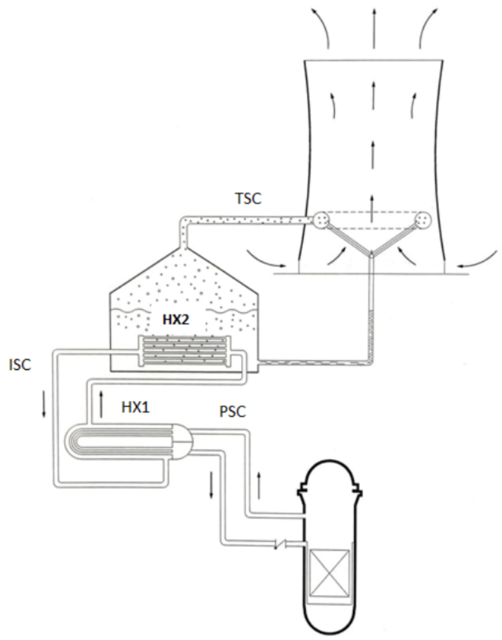

Figure 2.1: Passive Decay Heat Removal System of a fission reactors.

2.1.1 Input data and assumptions

To simplify the analysis of the Decay Heat Removal System (DHRS), a set of modeling assumptions is introduced. These assumptions allow the complex thermal–hydraulic behavior of the system to be described using a tractable analytical approach, while retaining the main physical mechanisms governing the natural circulation loops and the heat transfer processes in the heat exchangers.
The input data used for the calculations are summarized in 2.3.2.

(i) **Steady state**: The system is in steady state; the initial transient following reactor shutdown is neglected. The total decay heat to be dissipated is set to 1% of the plant's nominal thermal power ($P_{nom} = 600$ MW$_{th}$), resulting in a target heat removal capacity of 6 MW$_{th}$.

(ii) **Cooling tower capacity**: The cooling tower exhausts all the power necessary to keep the pool at saturation conditions.

(iii) **Constant pressure**: Constant coolant pressure in the three circuits to compute water and steam properties.

(iv) **Negligible fouling**: No fouling resistance in the heat exchangers.

(v) **LMTD correction factor**: Correction factor for the mean log temperature difference calculations (HX1 is not a pure countercurrent heat exchanger) equal to 0.7 in HX1.

(vi) **Equivalent manifold value**: The manifold contains several tubes with different dimensions. To simplify the analysis, the system is modeled using an equivalent single flow passage whose cross-sectional area is equal to the sum of the areas of the individual tubes. For a more rigorous estimation, the localized loss coefficients should be derived using the correlations provided in Idelchik's *Handbook of Hydraulic Resistance* [2].

(vii) **Constant concentrated loss coefficient in the core:** For simplicity, the core pressure drop is modeled using a constant lumped resistance term. This parameter is evaluated based on the nominal operating conditions, knowing that the pressure drop is $\Delta p_{core} = 1.2$ bar at a mass flow rate of $\dot{m} = 3200$ kg/s. It should be noted that this approximation does not account for fluid density variations occurring at different flow rates.

(viii) **Constant pool temperature:** The water in the pool heat exchanger is assumed to be at saturation conditions at atmospheric pressure (1 bar).

2.1.2 Goals

- Compute the steady-state mass flow rates and the fluid temperature in both the Primary System Circuit (PSC) and the Intermediate System Circuit (ISC).

- Determine the coolant temperatures across the heat exchangers (HX1 and HX2) to verify efficient heat transfer.

- Provide a detailed breakdown of pressure drops within the PSC and ISC, summarized in dedicated tables, to ensure that the buoyancy head is sufficient to overcome all frictional and localized losses.

2.2 Mathematics

This section describes the mathematical model and the correlations adopted to verify the DHRS performance. The problem involves a coupled thermal-hydraulic analysis of the Primary System Circuit (PSC) and the Intermediate System Circuit (ISC).

2.2.1 Equations

The steady-state behavior of each natural circulation loop is governed by the conservation of mass, momentum, and energy.

Mass and Energy Conservation

For each circuit, the mass flow rate $\dot{m}$ is constant along the loop. The heat power $\dot{Q}$ can be related to both the temperature difference between the hot (T_H) and cold (T_C) legs or the logarithmic mean temperature difference (ΔT_{lm}):

$$\dot{Q} = \dot{m} c_p \Delta T \quad \text{and} \quad \dot{Q} = UA\Delta T_{lm} F \tag{2.1}$$

where c_p is the specific heat capacity evaluated at the average temperature $T_{av} = (T_H + T_C)/2$. The LMDT is defined as:

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \tag{2.2}$$

where $\Delta T_1 = T_h - T_{h,ext}$ and $\Delta T_2 = T_c - T_{c,ext}$. By inverting Eq. 2.2, it is possible to analytically solve for the temperature in the hot and cold legs:

$$T_h = \frac{T_{h,ext} - (\Delta T + T_{c,ext}) \exp\left(\frac{\Delta T - (T_{h,ext} - T_{c,ext})}{\Delta T_{lm}}\right)}{1 - \exp\left(\frac{\Delta T - (T_{h,ext} - T_{c,ext})}{\Delta T_{lm}}\right)} \quad \text{and} \quad T_c = \Delta T - T_h \tag{2.3}$$

Finally, the overall heat transfer coefficient U for both heat exchangers is evaluated as follows:

$$\frac{1}{U} = \frac{1}{h_{ext}} + \frac{D_{out} \ln\left(\frac{D_{out}}{D_{in}}\right)}{2k_{th,tube}} + \frac{D_{out}}{h_{in} D_{in}} \tag{2.4}$$

Since U is formulated according to Eq. 2.4, the overall heat transfer and the resulting heat flux are intrinsically referred to the external surface area of the tubes.

Momentum Balance

The fundamental driving force is the natural circulation, where the buoyancy head Δp_B must equal the total pressure losses Δp_{loss}:

$$\Delta p_B = \Delta p_{loss} \tag{2.5}$$

As derived in the iterative algorithm, the pressure losses can be expressed as a function of the square of the mass flow rate using a loss geometry factor Π:

$$\Delta p_{loss} = \Pi \dot{m}^2 \implies \dot{m} = \sqrt{\frac{\Delta p_B}{\Pi}} \tag{2.6}$$

The term Π accounts for both distributed and concentrated losses across the loop:

$$\Pi = \sum_j \frac{f_j L_j}{D_j} \frac{1}{2\rho_j A_j^2} + \sum_l \frac{k_l}{2\rho_l A_l^2} \tag{2.7}$$

2.2.2 Parameters and Correlations

Since both heat exchangers are tube-based, the tube-side convective heat transfer coefficient is evaluated using the Gnielinski [3] correlation:

$$Nu = \frac{\frac{f}{8}(Re - 1000)Pr}{1 + 12.7\left(\frac{f}{8}\right)^{1/2}(Pr^{2/3} - 1)} \tag{2.8}$$

where f is the friction factor, Re is the Reynolds number inside the single tube, and Pr is the Prandtl number of the fluid. The internal heat transfer coefficient is subsequently computed as:

$$h_{in} = \frac{Nu\, k_{th,fluid}}{D_{in}} \tag{2.9}$$

Furthermore, the modeling of the two circuits requires specific correlations for heat transfer coefficients of the external area and pressure drops. The shell-side parameters are presented in the Appendix B.1.

Intermediate System Circuit (ISC)

The ISC is coupled to the pool at saturation temperature $T_{sat,pool}$.

- **Buoyancy Head:** Driven by the density difference over the net elevation H_{ISC}:

$$\Delta p_{B,ISC} = (\rho_c - \rho_h) g H_{ISC} \tag{2.10}$$

- **Pool Heat Transfer (McAdams):** On the pool side of HX2, the heat flux q'' is calculated as:

$$q'' = 2.257 \Delta T_{sat}^{3.86} \implies h_{pool} = \frac{q''}{\Delta T_{sat}} \tag{2.11}$$

  where $\Delta T_{sat} = T_{wall} - T_{sat,pool}$.

- **HX1 Shell-Side local loss coefficient**: For the pressure losses on the shell side, the local loss coefficient can be computed as:

$$k_{shell} = 8 \frac{0.227}{Re_{shell}^{0.193}} \frac{D_{shell}}{D_{eq,shell}} (n_{baffle} + 1) \tag{2.12}$$

Primary System Circuit (PSC)

The PSC involves a variable density distribution within the reactor core.

- **Buoyancy Head:** Composed of the contribution from the legs (H_1) and the core (H_2). Assuming a linear density variation in the core:

$$\Delta p_{B,core} = g \int_0^{H_2} (\rho_c - \rho(z)) dz = g \frac{\rho_c - \rho_h}{2} H_2 \tag{2.13}$$

  The total buoyancy head is $\Delta p_{B,tot} = (\rho_c - \rho_h) g H_1 + \Delta p_{B,core}$.

- **HX1 Shell-Side Heat Transfer:** The convective coefficient h_{shell} is derived from:

$$h_{shell} = 0.351 Re_{shell}^{0.55} \frac{k_{th,fluid}}{D_{eq,shell}} Pr^{1/3} \tag{2.14}$$

Friction and Minor Losses

The friction factor f is computed using the Haaland equation to account for pipe roughness ϵ:

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\left(\frac{\epsilon / D}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right] \tag{2.15}$$

Localized losses include sudden expansions and contractions at the headers:

$$k_{enl} = \left(1 - \frac{A_1}{A_2} \right)^2, \quad k_{contr} = 0.5 \left(1 - \frac{A_2}{A_1} \right) \tag{2.16}$$

When evaluating these localized losses, the loss coefficients are referred to the narrower cross-sectional area between the upstream and downstream pipes. In accordance with Munson in *Fundamentals of Fluid Mechanics* [4] , this area is chosen because it corresponds to the highest fluid velocity and, consequently, the maximum level of turbulence and energy dissipation.

2.3 Solution

Determining the mass flow rate requires an iterative approach. Since it represents the primary unknown of the system, several key parameters (such as the Reynolds number, the friction factor, and the overall heat transfer coefficient) are inherently dependent on it. As with the previous calculation, the numerical solver is developed in Python. The iterative algorithm is set to stop when the relative error becomes reaches a threshold ($\|m_{new} - m_{old}\| / m_{old} < 10^{-6}$).

2.3.1 Method Implemented

The algorithm is structured around a unified iterative loop, which is adaptable to both systems through the use of conditional statements and flags. Due to the presence of HX1, the PSC and the ISC are thermodynamically coupled. This interdependence manifests specifically in two aspects:

- Shell-side heat transfer coefficient: The convective coefficient h depends on the shell-side Reynolds number Re and Prandtl number Pr. This value is essential to evaluate the overall heat transfer coefficient of HX1, which subsequently governs the LMTD of the PSC.

- PSC leg temperatures: The LMTD of the primary circuit is a direct function of the cold and hot leg temperatures of the ISC.

For these reasons, the analysis begins by solving the ISC loop, starting with an initial guess for the mass flow rate ($\dot{m}$ = 100 kg/s) and the average fluid temperature (T_{av} = 120 °C).

1. Hydrodynamic characterization: The fluid properties are evaluated at the assumed temperature, and the Reynolds numbers are computed for both the tube side and the shell side in order to determine the friction factors and the shell-side loss coefficient.

2. Thermal evaluation: The correlations presented in Section **??** allow for the calculation of the overall heat transfer coefficient and, subsequently, the Logarithmic Mean Temperature Difference of the heat exchanger.

3. Temperature update: Using the LMTD and the macroscopic energy balance, the algorithm extracts the updated temperatures for the hot and cold legs (T_h and T_c). It is important to note that, for this specific heat exchanger (HX2), the external fluid is a boiling pool; therefore, it is modeled at a constant saturation temperature (T_{pool}).

4. Momentum balance: With the newly computed temperatures, the actual fluid densities are evaluated. The algorithm calculates the driving buoyancy force and aggregates all distributed and localized pressure drops, including pipe friction, bends, sudden expansions/contractions, and shell-side resistance.

5. Variable update and convergence check: The momentum equation is solved to find the new mass flow rate ($\dot{m}_{new}$) that balances buoyancy and friction. The average temperature is also updated. Finally, the relative error between the new and old mass flow rates is checked against the chosen tolerance.

Once the ISC algorithm reaches convergence, the resulting shell-side parameters (Re_{shell}, k_{shell}, boundary temperatures) are stored and passed as known inputs to the Primary System (PSC) loop. The algorithm for the PSC follows the same fundamental iterative logic as the previous loop, but introduces the following deviations:

(I) Shell-side heat exchanger: The PSC does not calculate the shell-side hydrodynamics. Instead, it inherits the converged parameters from the ISC to evaluate the convective coefficient.

(II) HX1 temperature update: Instead of assuming an isothermal heat sink, the algorithm utilizes the variable hot and cold leg temperatures of the ISC as the thermal boundaries. Moreover, since the heat exchanger is not perfectly counter-current, the LMTD is penalized by a dedicated correction factor.

(III) Two-stage buoyancy evaluation: The algorithm adds a specific core term to the standard buoyancy, evaluated through the application of an integral method, which accounts for the linear density gradient of the fluid as it heats up inside the core.

(IV) Localized pressure drops: The primary system is characterized by unique components. This includes an assigned constant pressure drop for the core, a block valve, U-tube bends, and a double-stage cross-sectional transition (from the main pipe to the header, and from the header to the U-tubes).

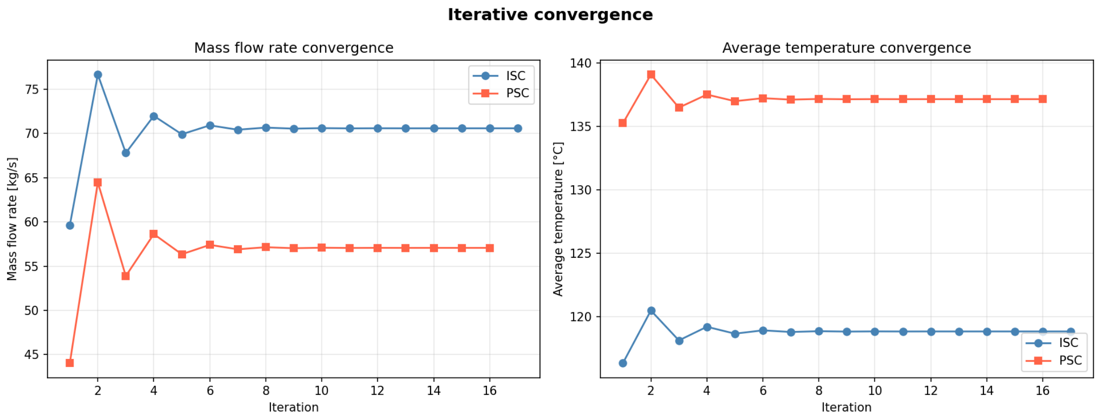

Figure 2.2: Convergence of mass flow and average temperature

2.3.2 Results

Following the convergence of the iterative solvers, the final data are presented in the tables below. Table 2.1 highlights the primary parameters governing the thermal-hydraulic design. Crucially, as the maximum temperature in the hot leg is consistently lower than the saturation temperature, the fluid is confirmed to remain strictly single-phase across the cycles.

Table 2.1: Results of design parameters for the ISC and PSC cycles.

Parameter	**ISC Cycle**	**PSC Cycle**
Mass flow rate	70.58 kg/s	57.30 kg/s
Cold leg temperature (T_c)	108.80 °C	124.82 °C
Hot leg temperature (T_h)	128.92 °C	149.43 °C
Saturation temperature	285.83 °C	290.54 °C

As shown in B.2, the external convective coefficients (h_{ext}) are lower than the internal ones (h_{in}) for both components. This indicates that the thermal resistance is predominantly governed by the external side (shell-side for HX1 and pool-boiling side for HX2). Consequently, the overall heat transfer coefficient U is heavily influenced by the external flow dynamics.

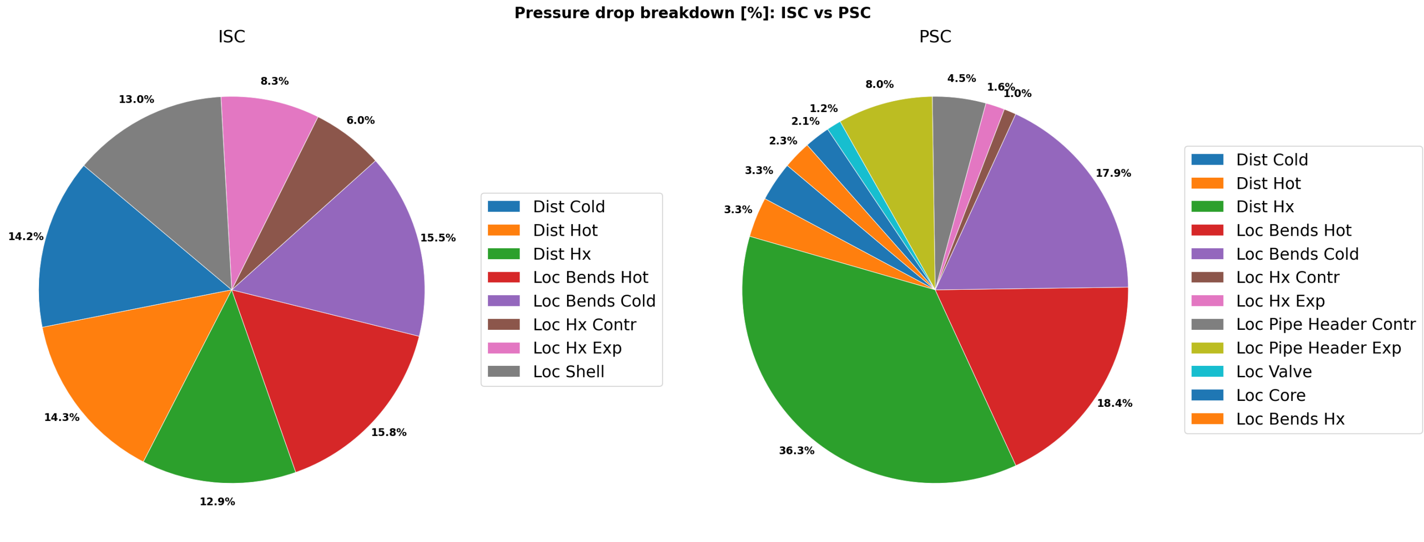

Figure 2.3: Percentage breakdown of the total pressure drop contributions for the PSC and ISC.

Tables 2.2 and 2.3 detail the contributions of individual components to the total pressure drop for the ISC and PSC, respectively. The data indicate that localized pressure drops dominate in both circuits, suggesting that geometric transitions and fittings (form losses) exert a greater hydraulic resistance than the distributed skin friction within the straight piping. However, the primary source of distributed losses differs between the two systems: in the ISC, the main piping accounts for the largest share, whereas in the PSC, the Heat Exchanger (HX1) is the predominant contributor. This is attributed to the HX1 design, which features a large bundle of small-diameter tubes; the reduced hydraulic diameter and consequently higher local fluid velocities lead to a significantly increased frictional resistance.

Table 2.2: Detailed breakdown of distributed and localized pressure drops for the ISC cycle.

Component	**Pressure Drop [Pa]**	**%**
Distributed Pressure Drops		
Cold leg pipe	222.24	14.2
Hot leg pipe	224.02	14.3
Heat Exchanger (tubes)	202.38	12.9
Total Distributed	**648.63**	
Localized Pressure Drops		
Bends (hot leg)	246.34	15.8
Bends (cold leg)	242.23	15.5
Heat Exchanger (contraction)	93.64	6.0
Heat Exchanger (expansion)	129.25	8.3
Shell side	203.17	13
Total Localized	**914.63**	

Table 2.3: Detailed breakdown of distributed and localized pressure drops for the PSC cycle.

Component	Pressure Drop [Pa]	%
Distributed Pressure Drops		
Cold leg pipe	59.08	3.3
Hot leg pipe	59.86	3.3
Heat Exchanger (tubes)	649.64	36.3
Total Distributed	**768.58**	
Localized Pressure Drops		
Bends (hot leg)	328.27	18.6
Bends (cold leg)	320.80	17.9
Pipe to header transition (contraction)	80.06	4.5
Header to pipe transition (expansion)	142.27	8.0
Header to tubes transition (contraction)	18.22	1.0
Tubes to header transition (expansion)	28.46	1.6
Valve	21.63	1.2
Core	38.16	2.1
U-tubes bends	42.00	2.3
Total Localized	**1019.88**	

Finally, as graphically summarized in Figure 2.4, it is essential to highlight the fundamental mechanism governing natural circulation. For both circuits, the sum of distributed and localized pressure drops is perfectly balanced by the buoyancy driving force at steady-state conditions. This exact equilibrium between total hydraulic resistance and driving head is what ultimately dictates the final mass flow rates established within the loops.

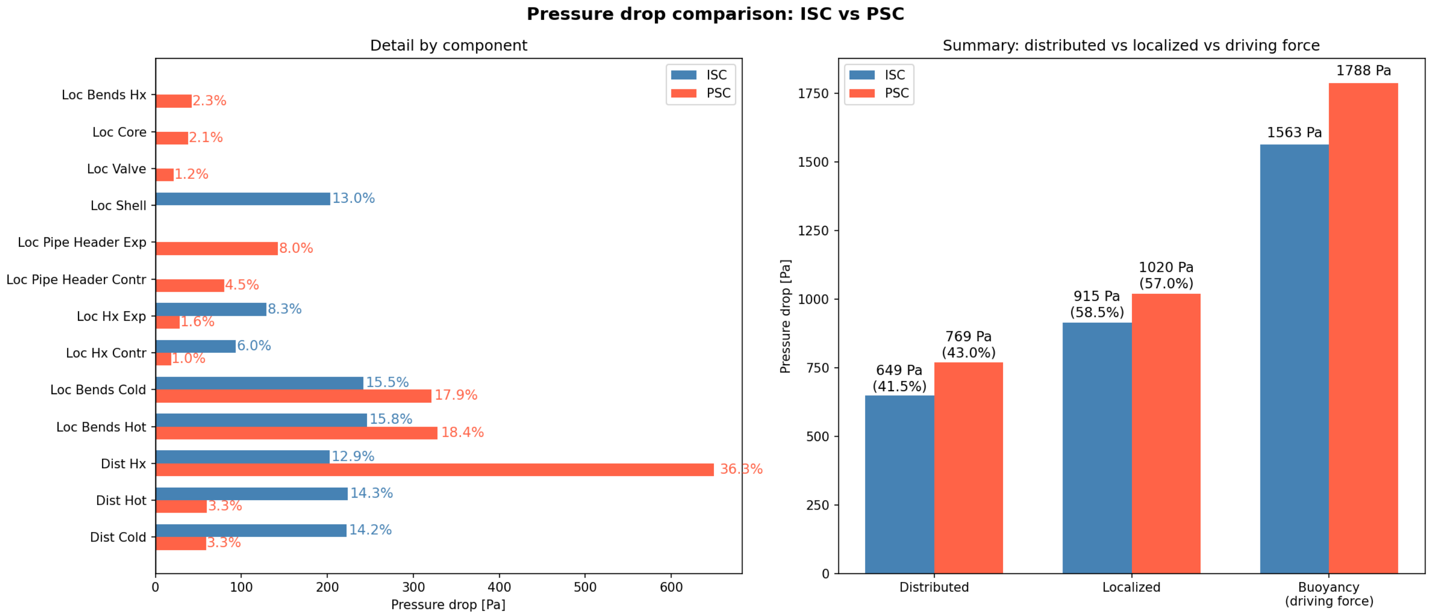

Figure 2.4: Comparison of pressure drops between the ISC and PSC.

Text reviewed and corrected with the support of Artificial Intelligence.

Appendix A: Problem design data

NPS (in)	Outside Diameter (in)	Schedule												
		10	20	30	STD	40	60	XS	80	100	120	140	160	XXS
		Wall Thickness (in)												
1/8"	0,405				0,068	0,068		0,095	0,095					
1/4"	0,540				0,088	0,088		0,119	0,119					
3/8"	0,675				0,091	0,091		0,126	0,126					
1/2"	0,840				0,109	0,109		0,147	0,147				0,187	0,294
3/4"	1,050				0,113	0,113		0,154	0,154				0,219	0,308
1"	1,315				0,133	0,133		0,179	0,179				0,250	0,358
1 1/4"	1,660				0,140	0,140		0,191	0,191				0,250	0,382
1 1/2"	1,900				0,145	0,145		0,200	0,200				0,281	0,400
2"	2,375				0,154	0,154		0,218	0,218				0,344	0,436
2 1/2"	2,875				0,203	0,203		0,276	0,276				0,375	0,552
3"	3,500				0,216	0,216		0,300	0,300				0,438	0,600
3 1/2"	4,000				0,226	0,226		0,318	0,318					
4"	4,500				0,237	0,237		0,337	0,337		0,438		0,531	0,674
5"	5,563				0,258	0,258		0,375	0,375		0,500		0,625	0,750
6"	6,625				0,280	0,280		0,432	0,432		0,562		0,719	0,864
8"	8,625		0,250	0,277	0,322	0,322	0,406	0,500	0,500	0,594	0,719	0,812	0,906	0,875
10"	10,750		0,250	0,307	0,365	0,365	0,500	0,500	0,594	0,719	0,844	1,000	1,125	1,000
12"	12,750		0,250	0,330	0,375	0,406	0,562	0,500	0,688	0,844	1,000	1,125	1,312	1,000
14"	14,000	0,250	0,312	0,375	0,375	0,438	0,594	0,500	0,750	0,938	1,094	1,250	1,406	
16"	16,000	0,250	0,312	0,375	0,375	0,500	0,656	0,500	0,844	1,031	1,219	1,438	1,594	
18"	18,000	0,250	0,312	0,438	0,375	0,562	0,750	0,500	0,938	1,156	1,375	1,562	1,781	
20"	20,000	0,250	0,375	0,500	0,375	0,594	0,812	0,500	1,031	1,281	1,500	1,750	1,969	
22"	22,000	0,250	0,375	0,500	0,375		0,875	0,500	1,125	1,375	1,625	1,875	2,125	
24"	24,000	0,250	0,375	0,562	0,375	0,688	0,969	0,500	1,219	1,531	1,812	2,062	2,344	
30"	30,000	0,312	0,500	0,625	0,375			0,500						
32"	32,000	0,312	0,500	0,625	0,375	0,688								
34"	34,000	0,312	0,500	0,625	0,375	0,688								
36"	36,000	0,312	0,500	0,625	0,375	0,750								
42"	42,000		0,500	0,625	0,375	0,750								

- STD - Standard
- XS - Extra Strong
- XXS - Double Extra Strong
- NPS - Nominal Pipe Size

Figure A.1: Pipe diameters from ANSI B36.10

Parameter	Symbol	Value	Units
Primary System Circuit (PSC)			
Coolant pressure	p	75	bar
Outer pipe diameter	D_{out}	16 (schedule 100)	in
Single pipe length	L	8	m
Pipe relative roughness	ϵ/D	$2 \cdot 10^{-4}$	-
Pressure loss coefficient (90° bend)	k_{loss}	0.45	-
Number of 90° bends per pipe	N_{bends}	4	-
Valve loss coefficient	$k_{loss,valve}$	0.12	-
Elevation hot–cold leg	H_1	7	m
Core elevation difference	H_2	3	m
Heat Exchanger 1 (HX1)			
Outer tube diameter	$D_{out,tube}$	19.05	mm
Tube thickness	t	1.24	mm
Number of tubes	N_t	897	-
Tube pitch	d_p	28.5	mm
Shell inner diameter	D_{shell}	1.5	m
Number of baffles	N_b	2	-
Baffles spacing	$l_{baffles}$	1.6	m
Average tube length	L_{tubes}	9.314	m
Tube thermal conductivity	k_{th}	15	W $m^{-1}K^{-1}$
Headers flow area	A_h	0.883	m^2
Heat transfer area	$A_{ht,HX1}$	500	m^2
Tube relative roughness	ϵ/D	10^{-4}	-
LMTD correction factor	F_T	0.7	-
Intermediate System Circuit (ISC)			
Coolant pressure	p	70	bar
Outer pipe diameter	$D_{out,tube}$	16 (schedule 100)	in
Single pipe length	L_{pipe}	20	m
Pressure loss coefficient (90° bend)	k_{loss}	0.45	-
Number of 90° bends per pipe	N_{bends}	2	-
Tube relative roughness	ϵ/D	$2 \cdot 10^{-4}$	-
Net elevation	H_{ISC}	10	m
Heat Exchanger 2 (HX2)			
Tube outer diameter	$D_{out,tube}$	25.4	mm
Tube thickness	t	1.24	mm
Number of tubes	N_t	770	-
Average tube length	L_{tubes}	7	m
Manifold diameter	$D_{manifold}$	16	in
Tube thermal conductivity	k_{th}	15	W $m^{-1}K^{-1}$
Heat transfer area	$A_{ht,HX2}$	430	m^2
Tube relative roughness	ϵ/D	10^{-4}	-

Table A.1: Geometrical and material input data

Appendix B: General reactor design parameters

B.1 Shell-side parameters

To evaluate the thermal-hydraulic performance on the shell side, the equivalent diameter ($D_{eq,shell}$) for a triangular tube pitch and the cross-flow area (A_{shell}) are computed as follows:

$$D_{eq,shell} = \frac{2\sqrt{3}d_p^2}{\pi D_{tube}} - D_{tube} \tag{B.1}$$

$$A_{shell} = \frac{D_{shell}}{d_p}(d_p - D_{tube})l_{baffles} \tag{B.2}$$

Based on these geometric characteristics, the shell-side Reynolds number (Re_{shell}) is directly evaluated as:

$$Re_{shell} = \frac{\dot{m}\, D_{eq,shell}}{A_{shell}\, \mu} \tag{B.3}$$

B.2 Heat Transfer Coefficients

Table B.1: Calculated heat transfer coefficients for HX1 and HX2. All values are expressed in $W/(m^2K)$.

Parameter	HX1	HX2
Internal convective coefficient (h_{in})	3337.9	2365.4
External convective coefficient (h_{ext})	1591.4	1453.8
Overall heat transfer coefficient (U)	942.2	804.3

Bibliography

[1] Samuele Meschini. *Nuclear Fission Plants - Assignment 1.2: Analysis of a Passive Decay Heat Removal System*. Course materials, Politecnico di Torino. Academic Year 2025/2026. 2025 (cit. on pp. 2, 10).

[2] Isaak E. Idelchik. *Handbook of Hydraulic Resistance*. 3rd. Begell House, 1994 (cit. on p. 11).

[3] Frank P. Incropera, Theodore L. Bergman, Adrienne S. Lavine, and David P. DeWitt. *Introduction to Heat Transfer*. 6th ed. Hoboken, NJ: John Wiley & Sons, 2011 (cit. on p. 13).

[4] Bruce R. Munson, Donald F. Young, Theodore H. Okiishi, and Wade W. Huebsch. *Fundamentals of Fluid Mechanics*. 7th. John Wiley & Sons, 2012 (cit. on p. 14).